

Flexible Bivariate Regression

September 1, 2022

Outline

Flexibility with “Linear” Regression

Flexibility with ML approaches

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Nonlinear “Linear” Regression

- Fitted Values from the LCMF $\hat{y}_i = B_0 + B_1 x_i + B_2 x_i^2$ for $i = 1, \dots, N$
- Residuals from Fitted Values: $e_i = y_i - \hat{y}_i$ for $i = 1, \dots, N$

Again, we choose B_0 , B_1 , and B_2 , to minimize the SSR, where

$$SSR = \sum_{i=1}^N e_i^2$$

but the definition of e_i changes with the inclusion of polynomial terms.

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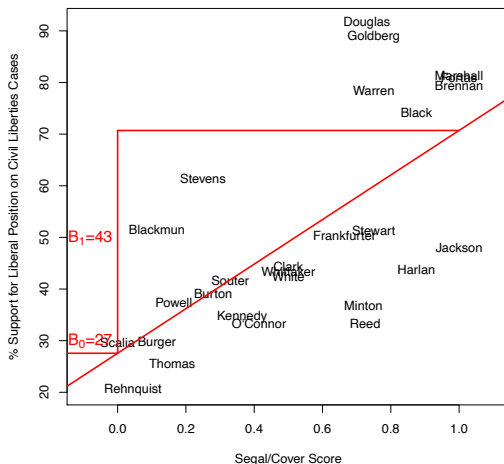
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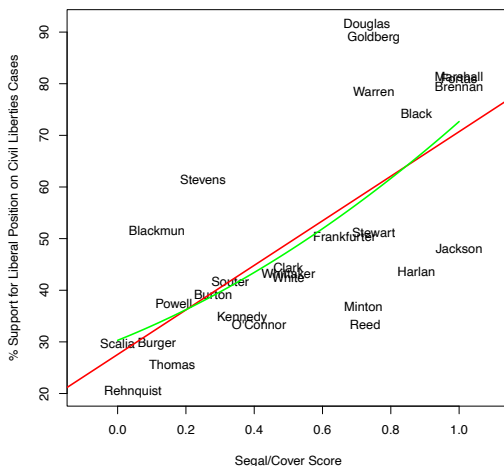
Recall: Simple Regression with SCscore

$$\widehat{CLib}_i = B_0 + B_1 \cdot SCscore_i$$



Quadratic Regression with SCscore

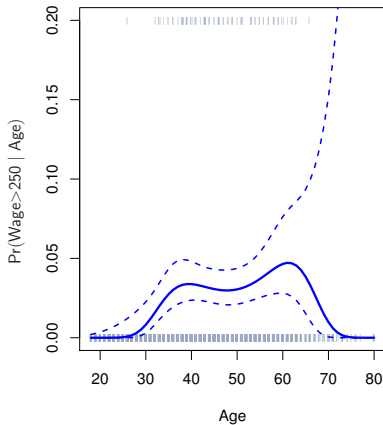
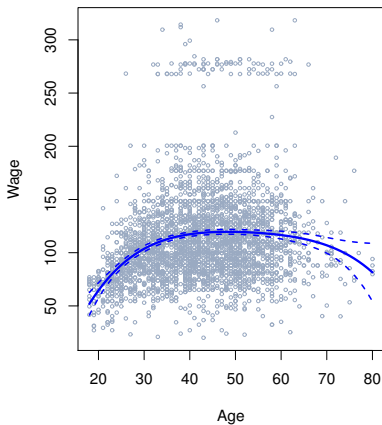
$$\widehat{CLib}_i = B_0 + B_1 \cdot SCscore_i + B_2 \cdot SCscore_i^2$$



Polynomial Regression

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \beta_3 x_i^3 + \dots + \beta_d x_i^d + \epsilon_i$$

Degree-4 Polynomial



Details

- Create new variables $X_1 = X$, $X_2 = X^2$, etc and then treat as multiple linear regression.
- Not really interested in the coefficients; more interested in the fitted function values at any value x_0 :

$$\hat{f}(x_0) = \hat{\beta}_0 + \hat{\beta}_1 x_0 + \hat{\beta}_2 x_0^2 + \hat{\beta}_3 x_0^3 + \hat{\beta}_4 x_0^4.$$

Details continued

- Logistic regression follows naturally. For example, in figure we model

$$\Pr(y_i > 250|x_i) = \frac{\exp(\beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \dots + \beta_d x_i^d)}{1 + \exp(\beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \dots + \beta_d x_i^d)}.$$

- To get confidence intervals, compute upper and lower bounds on *on the logit scale*, and then invert to get on probability scale.

Outline

Flexibility with “Linear” Regression

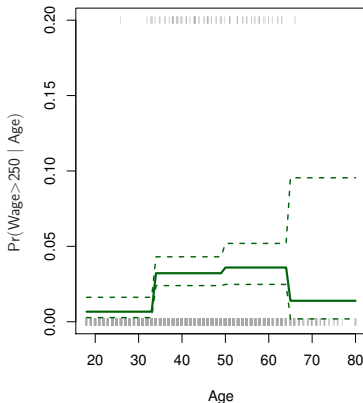
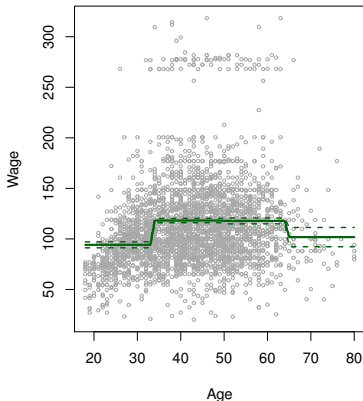
Flexibility with ML approaches

Step Functions

Another way of creating transformations of a variable — cut the variable into distinct regions.

$$C_1(X) = I(X < 35), \quad C_2(X) = I(35 \leq X < 65), \dots, C_3(X) = I(X \geq 65)$$

Piecewise Constant



Step functions continued

- Easy to work with. Creates a series of dummy variables representing each group.
- Useful way of creating interactions that are easy to interpret. For example, interaction effect of **Year** and **Age**:

$$I(\text{Year} < 2005) \cdot \text{Age}, \quad I(\text{Year} \geq 2005) \cdot \text{Age}$$

would allow for different linear functions in each age category.

- In R: `I(year < 2005)` or `cut(age, c(18, 25, 40, 65, 90))`.

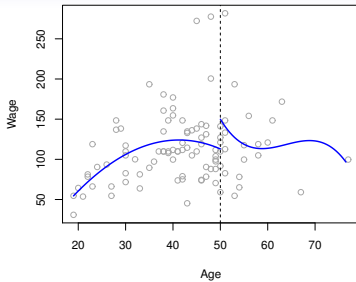
Piecewise Polynomials

- Instead of a single polynomial in X over its whole domain, we can rather use different polynomials in regions defined by knots. E.g. (see figure)

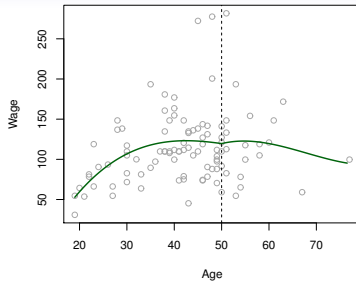
$$y_i = \begin{cases} \beta_{01} + \beta_{11}x_i + \beta_{21}x_i^2 + \beta_{31}x_i^3 + \epsilon_i & \text{if } x_i < c; \\ \beta_{02} + \beta_{12}x_i + \beta_{22}x_i^2 + \beta_{32}x_i^3 + \epsilon_i & \text{if } x_i \geq c. \end{cases}$$

- Better to add constraints to the polynomials, e.g. continuity.
- *Splines* have the “maximum” amount of continuity.

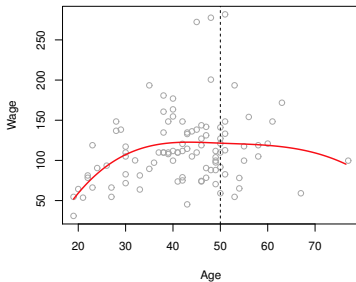
Piecewise Cubic



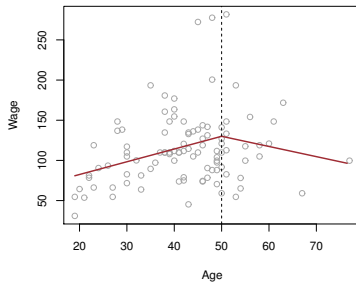
Continuous Piecewise Cubic



Cubic Spline

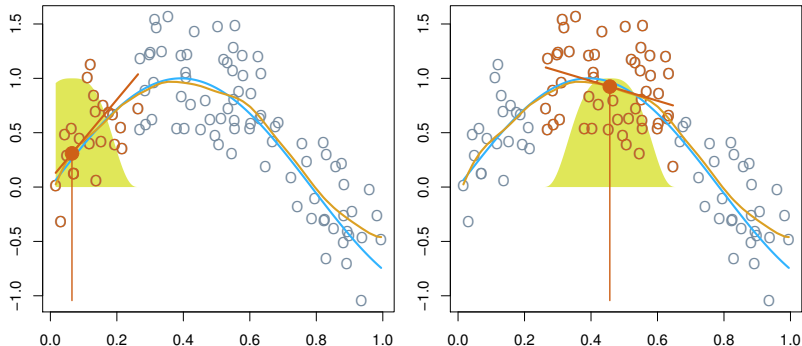


Linear Spline



Local Regression

Local Regression

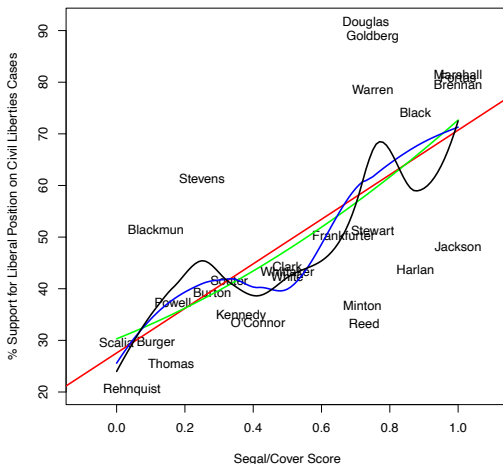


With a sliding weight function, we fit separate linear fits over the range of X by weighted least squares.

See text for more details, and `loess()` function in R.

Loess with SCscore

$$\widehat{CLib}_i = f(SCscore_i)$$



Summary and transition to board work

- We can increase flexibility within “linear” regression by adding polynomial terms.
- ML techniques like splines and kernel methods (e.g., loess) provide a better option
- On the board.... We can choose flexibility with LOOCV.